

A reproducible benchmark for transformation uncertainty in aquifer-test parameter inference

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ABSTRACT

Aquifer-test transmissivity and storage are not direct observations; they are inferred by fitting a response model to drawdown. This paper presents a reproducible computational benchmark for quantifying how that drawdown-to-parameter transformation affects inferred aquifer properties before they enter groundwater models, pumping-capacity calculations, or response-time estimates. Four analytical interpretations—Theis confined flow, Hantush–Jacob leakage, lagging Darcy without leakage, and lagging Darcy with leakage—are fitted to controlled numerical responses and two field pumping tests. A 10,000 scenario groundwater-flow ensemble generated 160,012 fitted interpretation records across homogeneous, leaky, finite-boundary, heterogeneous, and combined settings. The benchmark provides support-matched reference parameters, analytical limiting checks, grid sensitivity checks, and scenario-conditioned model factors. Analytical verification confirmed physical consistency in 22 of 25 limiting cases, and numerical responses aligned with analytical references in 30 of 32 configurations. Benchmark-transfer regressions predicted the 90th percentile absolute log model factor with cross-validated R^2 values of 0.48, 0.51, and 0.55 for transmissivity, storage, and response time. Application to the Massachusetts fractured-bedrock and Lovelock Valley basin-fill cases produced benchmark-conditioned model-factor COV values of 0.61–0.90 and 0.31–0.51, respectively. Model-averaged decision diagnostics showed that the selected interpretation can understate hydraulic capacity and response-time spread. The workflow treats lagging Darcy as one candidate interpretation, not a universal replacement, and reports transformation uncertainty through response residuals, benchmark support, field applicability, and inherited decision consequences.

1. Introduction

Groundwater models, capture calculations, dewatering designs, geothermal screening, and water-management assessments commonly inherit transmissivity and storage from pumping-test reports (Freeze and Cherry, 1979; Bear, 1979; Anderson et al., 2015). Those quantities are not measured directly. They are inferred by transforming observed drawdown through an interpretation model, such as Theis confined flow, Hantush–Jacob leakage, delayed-yield, wellbore-storage, or generalized radial-flow solutions (Theis, 1935; Cooper and Jacob, 1946; Hantush and Jacob, 1955; Boulton, 1954; Papadopoulos and Cooper, 1967; Neuman, 1975; Barker, 1988). This front-end transformation can introduce model-form error before stochastic groundwater modeling, Bayesian model averaging, hydraulic tomography, or management optimization begins (Rojas et al., 2008; Tsai, 2010; Tsai and Elshall, 2013; Doherty, 2015).

The distinction matters because mature groundwater uncertainty methods usually operate after a model ensemble, parameterization, or prior structure has already been specified. Slug-test support studies, pumping-test upscaling, aquifer-scale inference, and multimodel groundwater prediction address related downstream questions (Beckie and Harvey, 2002; Manewell et al., 2023, 2024; Naderi and Gupta, 2023; Li and Tsai, 2009). The present problem occurs

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earlier: a drawdown record is converted into apparent transmissivity, storage, leakage, or response-time parameters. If the response model omits temporal memory, leakage, boundary behavior, or unresolved support effects, the mismatch can be hidden inside fitted aquifer parameters and then mistaken for hydrogeologic heterogeneity.

This paper develops a reproducible benchmark for that upstream transformation step. Four analytical interpretation pathways are used as controlled drawdown-to-parameter mappings: Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage. Lagging Darcy is treated as one diagnostic interpretation for temporal response structure, not as a universal replacement for conventional aquifer-test solutions. The computational contribution is a 10,000 scenario MODFLOW 6 benchmark that provides controlled transient responses, support-matched reference parameters, analytical limiting checks, grid-sensitivity checks, and scenario-conditioned model factors. The benchmark is then linked to two public field data sets—a Massachusetts fractured-bedrock test and the Lovelock Valley MWP2 basin-fill test—through response diagnostics and model-averaged decision quantities (Cho et al., 2004, 2008; Nadler, 2020a,b).

2. Methods

2.1. Transformation uncertainty framework

The analysis treats aquifer-test interpretation as a transformation model. The measured input is a drawdown response $s_{\text{obs}}(t, r)$, and the derived outputs are apparent aquifer parameters θ_M , including transmissivity T , storage S , leakage factor B , or lagging response times under interpretation model M . For a fitted response $s_M(t, r; \theta_M)$, the response-level log model factor is

$$\eta_{M,i,j} = \log[s_{\text{obs},i}(t_j) + c] - \log[s_M(t_j; \theta_M) + c], \quad (1)$$

where i denotes an observation well, j denotes time, and c is a small positive offset used only to keep the log transformation defined. The dispersion SD_e of η provides a response-level measure of transformation residual structure. For numerical benchmark cases with known support-matched reference values, parameter model factors are also computed, for example $M_T = T_{\text{app}}/T_{\text{ref}}$ and $M_S = S_{\text{app}}/S_{\text{ref}}$. Lognormal coefficient of variation values are computed from the standard deviation of $\log M$.

2.2. Field data and analytical interpretations

The field component uses two public pumping-test data sets with contrasting hydrogeologic settings and observation geometries. The Massachusetts case is a fractured crystalline bedrock pumping test with 12 retained observation wells and 420 drawdown observations (Cho et al., 2004, 2008). The retained diagnostic subset is analyzed as a constant-

rate response with $Q = 1.2 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$, while the original variable-rate history remains an applicability condition. The Lovelock Valley case is USGS multiwell pumping test 2 in basin-fill sediments of the lower Humboldt River Basin, with pumping and recovery represented through superposition (Nadler, 2020a,b).

Both field cases are fitted using four analytical interpretations. Theis confined flow provides the classical no-leakage reference. Hantush–Jacob provides the conventional leaky-aquifer reference. Lagging Darcy without leakage introduces flux and storage-release lag times to diagnose temporal memory, and lagging Darcy with leakage combines that temporal response with leakage. The lagging solutions are evaluated in the Laplace domain and inverted using the Stehfest algorithm (Stehfest, 1970). Fits minimize log-response residuals using the same well coordinates, observation records, and pumping representation for all interpretations. AIC, BIC, local parameter correlation, boundary-hit checks, and time-window residual summaries are retained as complexity and identifiability diagnostics.

2.3. Numerical groundwater-flow benchmark

The controlled benchmark uses MODFLOW 6 as a reproducible groundwater-flow response generator (Langevin et al., 2017; Bakker et al., 2016). MODFLOW 6 is not the methodological focus; it supplies transient drawdown responses for testing how analytical interpretations recover apparent aquifer parameters. The ensemble contains 10,000 scenarios in ten classes: homogeneous confined flow, distributed leakage, finite no-flow boundary, constant-head boundary, heterogeneous hydraulic conductivity, anisotropic heterogeneous conductivity, heterogeneous storage, heterogeneous leakage, heterogeneous boundary, and combined leakage and boundary settings. Each class contributes 1000 scenarios.

The design spans grid spacings of 5.0–10.0 m, square domains of 405–1205 m, six observation layouts with 3–5 wells, logarithmically spaced records over 1–10 days, and water-level noise from 0 to 0.01 m. Heterogeneous cases use correlated lognormal hydraulic conductivity or storage fields. A scenario-specific effective pumping support radius defines T_{ref} and S_{ref} before fitting. The support radius combines hydraulic diffusivity, pumping duration, maximum observation distance, leakage length where present, and boundary or domain limits. This prevents a numerical cell value from being treated as the same object as a pumping-test parameter.

Each numerical response is refitted with the same four analytical interpretations. The resulting records are screened by removing active boundary-flag fits and retaining interpretations within $\Delta BIC \leq 10$ of the best interpretation for the same response. Analytical limit checks test whether the implemented formulas recover known limiting behavior. Numerical consistency checks compare MODFLOW-generated homogeneous and leaky responses against analytical reference responses across grid spacings.

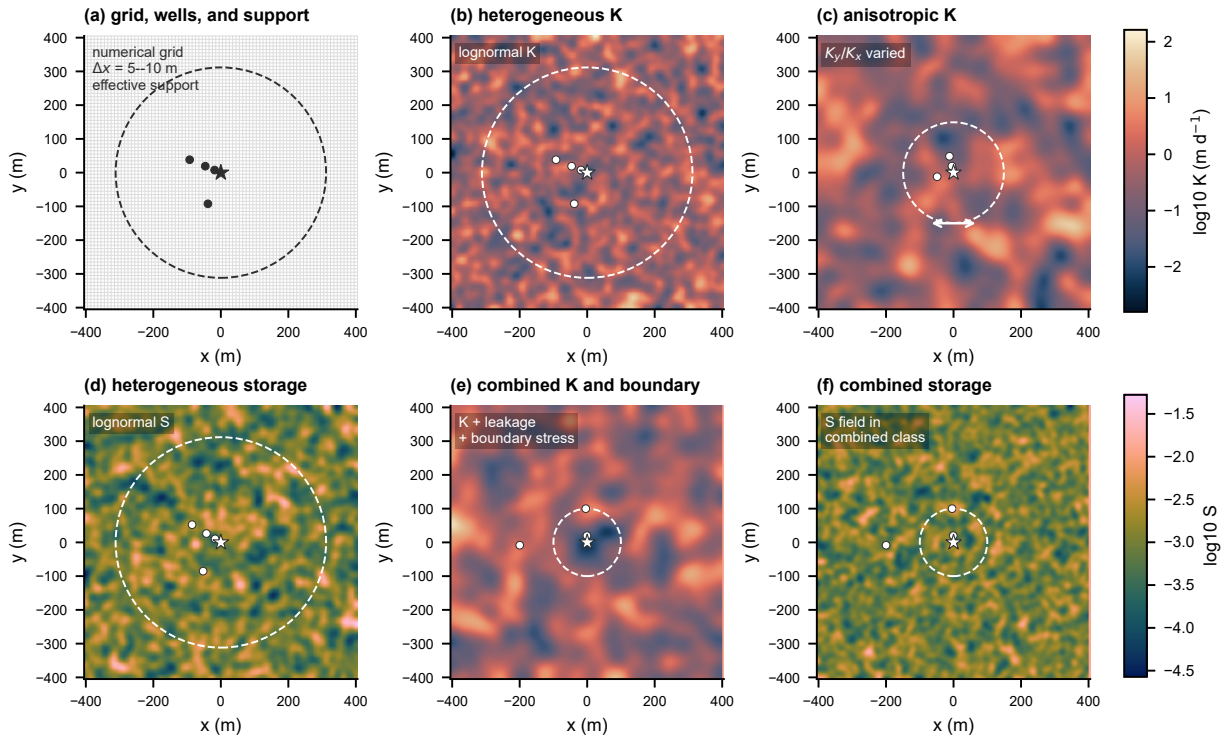


Figure 1: Numerical groundwater-flow benchmark design with MODFLOW 6 through (a) the finite-difference grid, extraction well, observation wells, and scenario-specific effective support envelope for the aquifer-test reference; (b) a representative heterogeneous hydraulic conductivity field; (c) a representative anisotropic hydraulic conductivity field; (d) a representative heterogeneous storage field; (e) combined leakage, boundary, hydraulic conductivity, and storage conditions for transformation stress; and (f) paired numerical response classes for the four analytical interpretations. The numerical fields define controlled drawdown responses for interpretation by the four analytical solutions.

2.4. Benchmark transfer and decision diagnostic

The numerical benchmark provides controlled reference values; the field cases provide real response diagnostics. Benchmark-transfer regressions therefore estimate the magnitude of parameter model factors that a field test may inherit when its diagnostics resemble parts of the numerical ensemble. Predictors are restricted to quantities available in both numerical and field tests: response residual dispersion, observation distance, BIC support, boundary and leakage flags, parameter spread, and fitted pathway class. Regression targets are the 90th percentile absolute log model factors for T , S , and response time within each numerical scenario.

The management diagnostic follows the Bayesian model averaging logic used in groundwater uncertainty analysis (Hoeting et al., 1999; Li and Tsai, 2009; Tsai and Elshall, 2013). The four aquifer-test interpretations are treated as candidate models. Their fitted T and S values are propagated into hydraulic capacity, proportional to T , and characteristic response time, proportional to S/T . Strict BIC weights and a variance-window weighting scheme are reported. This diagnostic asks whether a later groundwater calculation can inherit overconfident inputs if aquifer-test

Table 1

Numerical groundwater-flow benchmark for field-inference scope via MODFLOW 6 as the controlled response generator, scenario-conditioned transformation model factors for the four analytical interpretations, and field conditions under which benchmark factors can inform field diagnostics.

Component	Numerical benchmark result	Field interpretation
Coverage	10,000 numerical scenarios in ten classes; six observation layouts; 160,012 fitted interpretation and well records; 88,106 screened records.	Provides controlled response generation before field evidence is interpreted.
Verification	Analytical limit verification confirmed physical consistency in 22 of 25 cases; numerical consistency checks aligned with analytical reference responses in 30 of 32 cases; remaining warnings identify near well leakage and coarse grid limits.	Numerical support and discretization effects remain visible and should be reported.
Model factors	Screened class and interpretation median COV values are 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time; upper tails remain scenario dependent.	Conditional model factor COV values apply to matching benchmark classes.
Applicability	Massachusetts receives a primary conservative match for combined leakage and boundary behavior; Lovelock receives primary matches for heterogeneous storage and hydraulic conductivity.	Field geometry and true support scale define the applicable range of the scenario-conditioned values.

interpretation uncertainty is collapsed into one fitted parameter set.

3. Results and Discussion

3.1. Benchmark verification and model factors

The numerical benchmark generated 160,012 fitted interpretation and well records, of which 88,106 satisfied the screening criteria (Table 1). Analytical verification confirmed physical consistency in 22 of 25 limiting cases, and numerical responses aligned with analytical reference responses in 30 of 32 grid configurations (Figure 2). The remaining warnings identify near-well finite leakage and coarse-grid settings that require caution before field transfer. These checks define the usable benchmark domain rather than serving as a general claim about every pumping-test condition.

The screened benchmark separates response fit from parameter recovery. Across class and interpretation groups, the median model-factor COV values are 2.06 for transmissivity, 1.27 for storage, and 1.21 for response time (Figure 3). Leaky, heterogeneous leakage, and combined boundary classes have wider upper tails. Variance attribution shows that response residual standard deviation is driven mainly by water-level noise and observation support, whereas transmissivity and storage model factors remain strongly interpretation dependent. This result supports the central premise: a small drawdown residual does not guarantee that the inferred T or S is close to its support-matched reference value.

3.2. Benchmark-conditioned transfer to field diagnostics

Benchmark-transfer regressions predicted the 90th percentile absolute log model factor with cross-validated R^2 values of 0.48 for transmissivity, 0.51 for storage, and 0.55 for response time (Figure 4). The values are moderate, not definitive, but they provide a reproducible bridge from response diagnostics to conditional parameter uncertainty. Applied to the field cases, the p90-equivalent benchmark-transferred model-factor COV values are 0.61–0.90 for Mas-

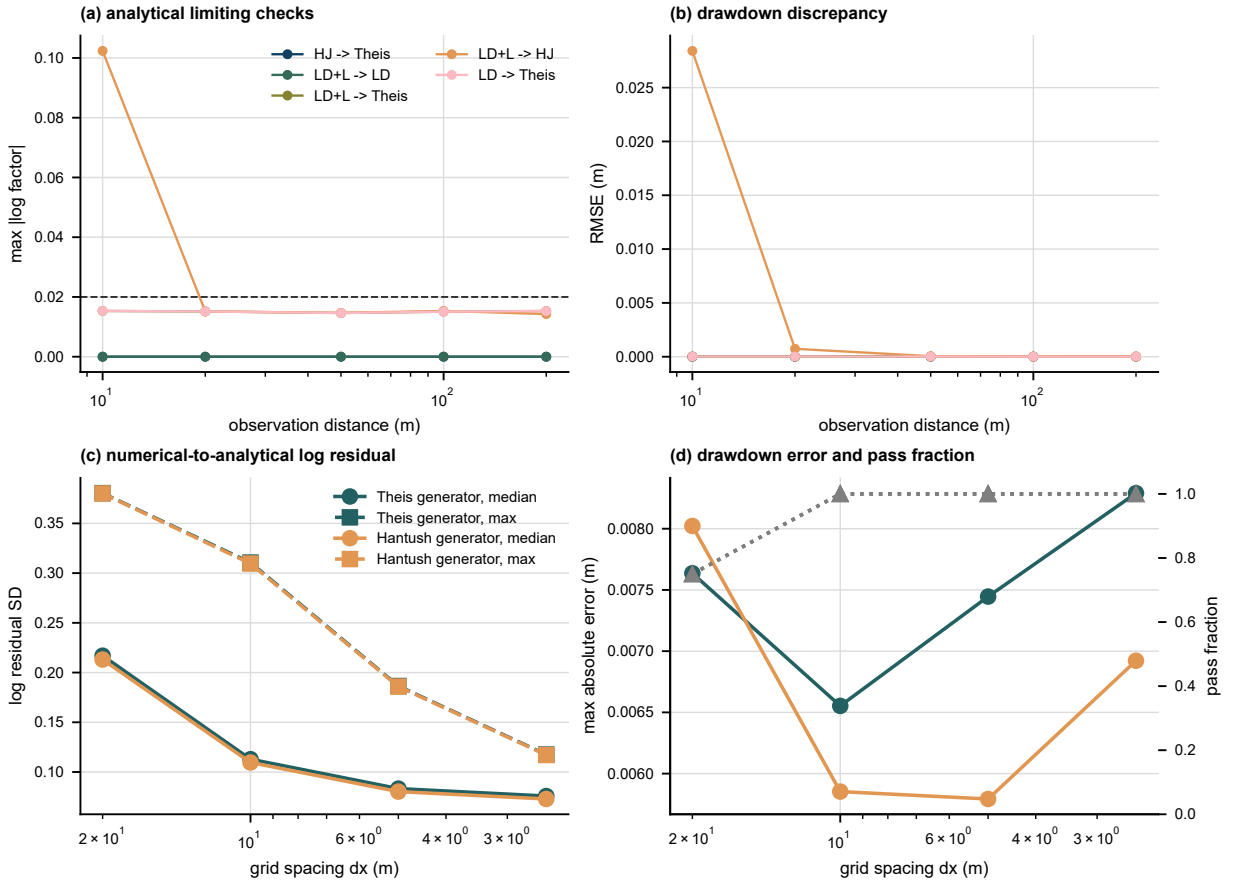


Figure 2: Analytical verification and numerical consistency checks through (a) analytical limit behavior versus observation distance and the dashed log factor tolerance for limit checks; (b) additional analytical limit behavior versus observation distance; (c) numerical consistency with homogeneous Theis analytical reference responses across grid intervals; and (d) numerical consistency with vertical leakage Hantush analytical reference responses across grid intervals. MODFLOW 6 numerical responses and caution labels define finite leakage, near-well, and coarse-grid limits for benchmark scope.

sachusetts and 0.31–0.51 for Lovelock Valley. These values are not universal design factors; they are sensitivity brackets conditioned on the represented benchmark classes and field diagnostics.

3.3. Field diagnostic evidence

The two field cases place the benchmark in real aquifer-test settings with different geological supports (Figure 5). Massachusetts is a small fractured-bedrock network, whereas Lovelock Valley is a basin-fill system with pumping and recovery observations. The four analytical interpretations were fitted consistently across both cases (Figures 6 and 8). The visual fits and residual summaries show that the lagging interpretations reduce structured response mismatch in both settings, while leakage support differs by case and well.

In Massachusetts, pooled log model factor dispersion decreases from 0.166 for Theis confined to 0.062 for lagging Darcy without leakage and 0.035 for lagging Darcy with leakage (Figure 7). Hantush–Jacob and lagging leaky forms

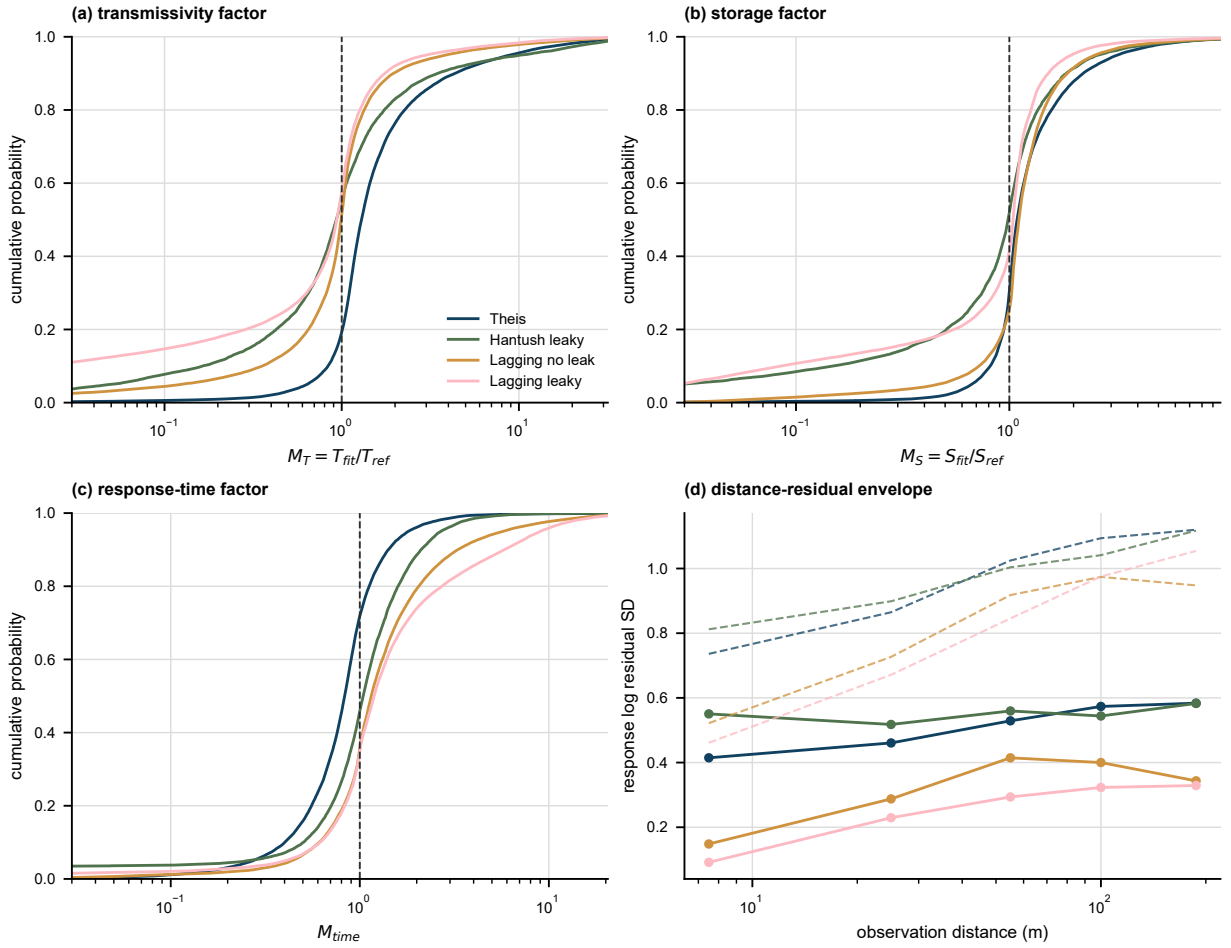


Figure 3: Screened numerical benchmark model factors through (a) empirical distribution functions for transmissivity model factors after boundary-active fit removal and retention of interpretations within $\Delta BIC \leq 10$; (b) empirical distribution functions for storage model factors under the same screened records; (c) empirical distribution functions for response-time model factors under the same screened records; and (d) median response log residual standard deviation by observation distance, with dashed lines for the 90th percentile envelope. The vertical dashed line in (a)–(c) denotes a factor of one, and all values represent scenario-conditioned model factors for the benchmark classes.

receive leakage support in selected wells, indicating that leakage or boundary-equivalent behavior should remain an explicit competing interpretation. In Lovelock Valley, pooled dispersion decreases from 0.246 for Theis confined to approximately 0.174 for the two lagging interpretations, while BIC support concentrates in the no-leakage lagging form (Figure 9). The site contrast supports the intended use of the four pathways: conventional leakage, temporal memory, and combined behavior are tested rather than assumed.

3.4. Scope of inference and management consequence

The synthesis figure separates what the two field cases can and cannot support (Figure 10). Both cases support response-level diagnostics and pathway-dependent parameter spread. The Massachusetts case provides a within-

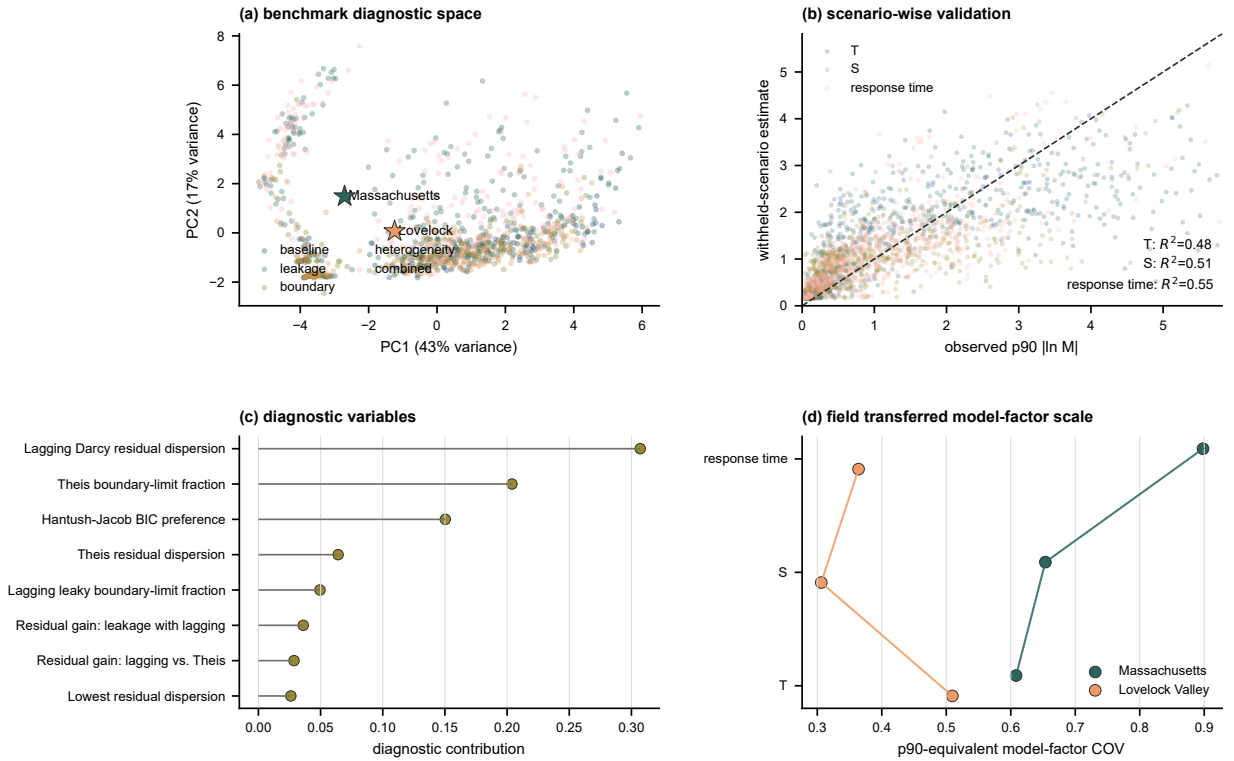


Figure 4: Benchmark transfer of aquifer-test transformation uncertainty to field diagnostics through (a) the numerical benchmark response-diagnostic space and the positions of the two field cases; (b) cross-validated predictions versus observed 90th percentile absolute log model factors for transmissivity, storage, and response time; (c) the main diagnostic variables for the regression; and (d) benchmark-transferred, 90th-percentile-equivalent model-factor COV values for the two field cases.

network spatial-transfer diagnostic by excluding one observation well at a time; Lovelock provides an independent field consistency check under a different support scale. Neither field case by itself recovers true support-scale T , S , leakage, or lagging times. The numerical benchmark supplies conditional model factors, and the field applicability criteria define where those factors can be used as sensitivity brackets or conservative priors (Figure 11).

The decision diagnostic shows why this upstream uncertainty matters for later groundwater calculations. Under variance-window BMA, the median robust hydraulic capacity factor is 1.60 in Massachusetts and 1.18 in Lovelock Valley; the corresponding 95th percentile factors are 2.73 and 4.52 (Figure 12). Median response-time factors are 1.85 and 4.05, with 95th percentile values of 3.32 and 8.79. These are diagnostic inheritance factors, not project-specific design factors. They indicate that groundwater calculations using a single interpreted T and S can become overconfident if the aquifer-test report does not state the interpretation model, residual structure, parameter spread, benchmark support, and field-to-benchmark applicability.

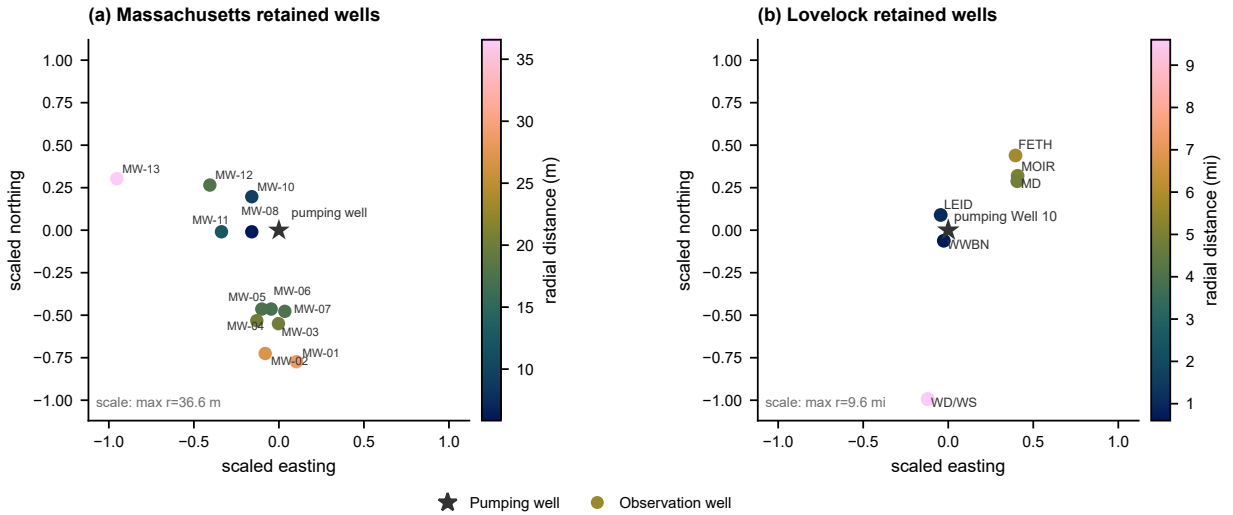


Figure 5: Paired field-data spatial support for the two real cases through (a) Massachusetts fractured bedrock retained observation wells; and (b) Lovelock Valley MWP2 retained observation wells relative to extraction Well 10. Both panels use coordinates centered on the extraction well and normalized by the maximum retained radial support within the case, so panel shape denotes observation geometry rather than absolute field scale. Circles denote retained observation wells, stars denote extraction wells, and colors retain the actual radial distance within each case.

3.5. Implications and limitations

The computational benchmark changes the role of field pumping tests in uncertainty analysis. Instead of treating reported transmissivity and storage as direct measurements, the workflow records the transformation that produced them. This makes it possible to distinguish response residuals, parameter model factors, and inherited decision factors. The distinction complements stochastic hydrogeology and groundwater BMA rather than replacing them: later models can only propagate uncertainty that is represented in their inputs.

The evidence also defines limits. The field COV values are specific to the two data sets and to the four implemented interpretations. Benchmark model factors are conditional on the scenario class, support definition, observation geometry, pumping schedule, grid design, and screening criteria. Direct evaluation of true field parameter recovery would require independent support-scale measurements. Broader model families—wellbore storage, skin, delayed yield, generalized radial flow, dual porosity, and unconfined numerical benchmarks—remain future extensions. Within these limits, the present result is a reproducible reporting standard: aquifer-test parameters should be accompanied by response model factors, complexity support, pathway-dependent parameter spread, benchmark applicability, and decision inheritance before they are used as deterministic groundwater model inputs.

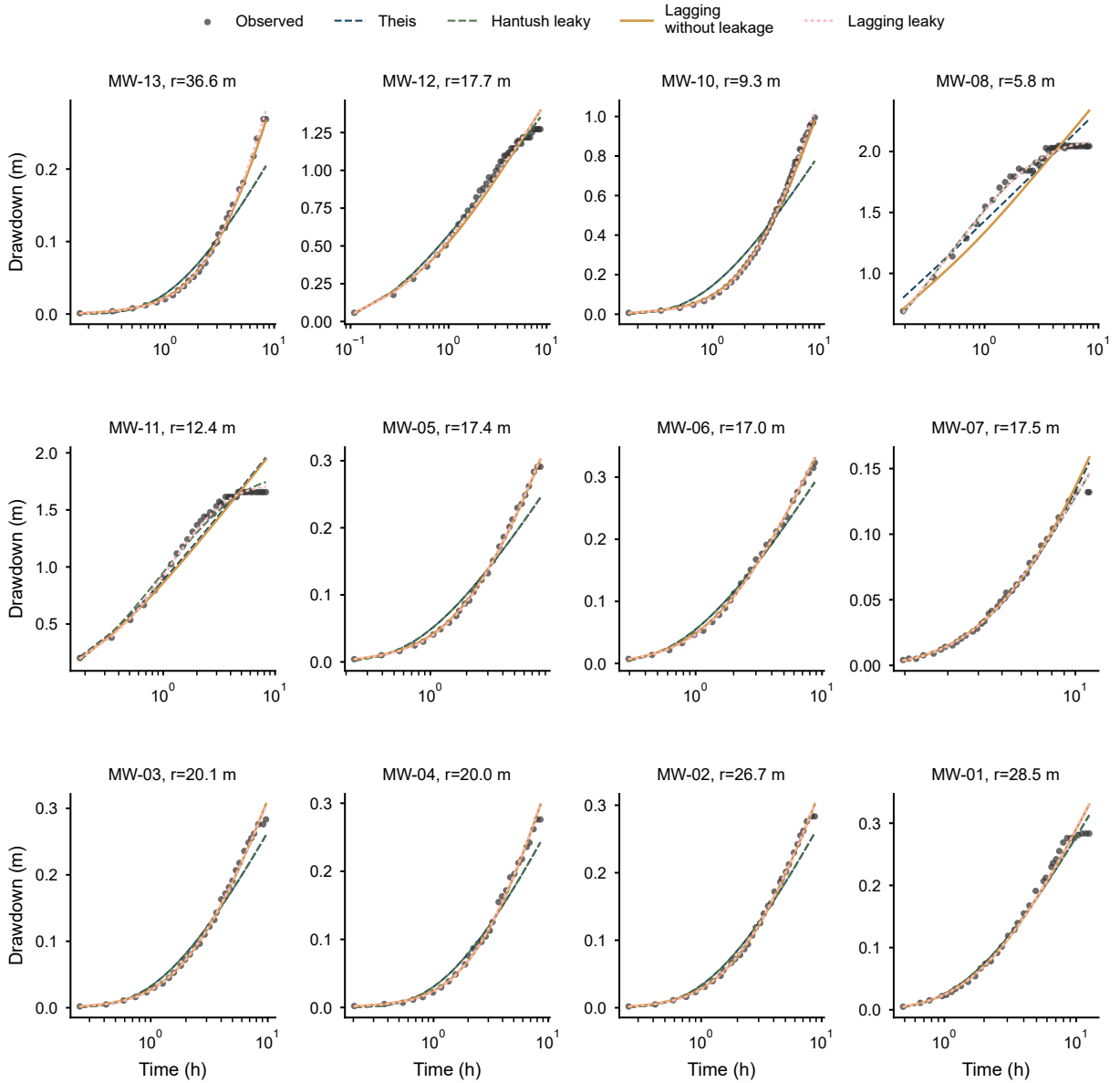


Figure 6: Massachusetts observed and simulated drawdown curves for the 12 observation wells across the four main interpretation models, with measured drawdown as points and Theis confined, Hantush–Jacob leaky, lagging Darcy without leakage, and lagging Darcy with leakage predictions as lines.

4. Conclusions

This study provides a CAGEO-oriented computational benchmark for aquifer-test transformation uncertainty. The workflow links measured drawdown, four analytical interpretation pathways, response-level log model factors, support-matched numerical reference values, benchmark model-factor distributions, field applicability criteria, and model-averaged decision diagnostics. The 10,000 scenario benchmark produced 160,012 fitted interpretation records, passed

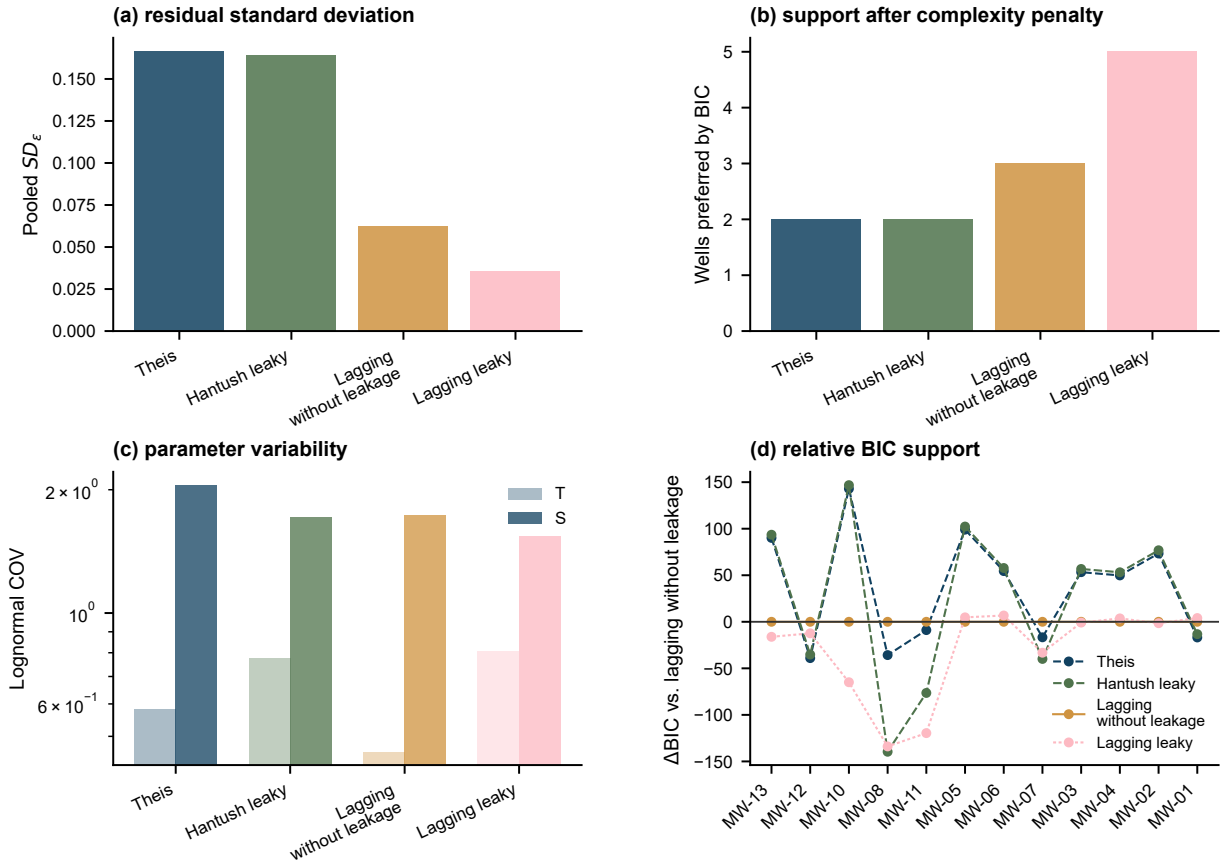


Figure 7: Massachusetts interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

22 of 25 analytical limiting checks and 30 of 32 numerical consistency checks, and showed that response fit, parameter recovery, and decision inheritance are distinct quantities.

The two field cases demonstrate the use of the benchmark. Lagging Darcy interpretations reduce response residual structure in both Massachusetts and Lovelock Valley, but leakage support, parameter convergence, and field transfer differ by case. The benchmark-transfer analysis gives conditional model-factor COV values rather than universal design values. The management diagnostic shows that collapsing interpretation uncertainty into one fitted parameter set can understate hydraulic capacity and response-time spread. The main contribution is therefore not a claim that one interpretation is universally superior; it is a reproducible code-backed workflow for reporting how aquifer-test drawdown becomes groundwater model input.

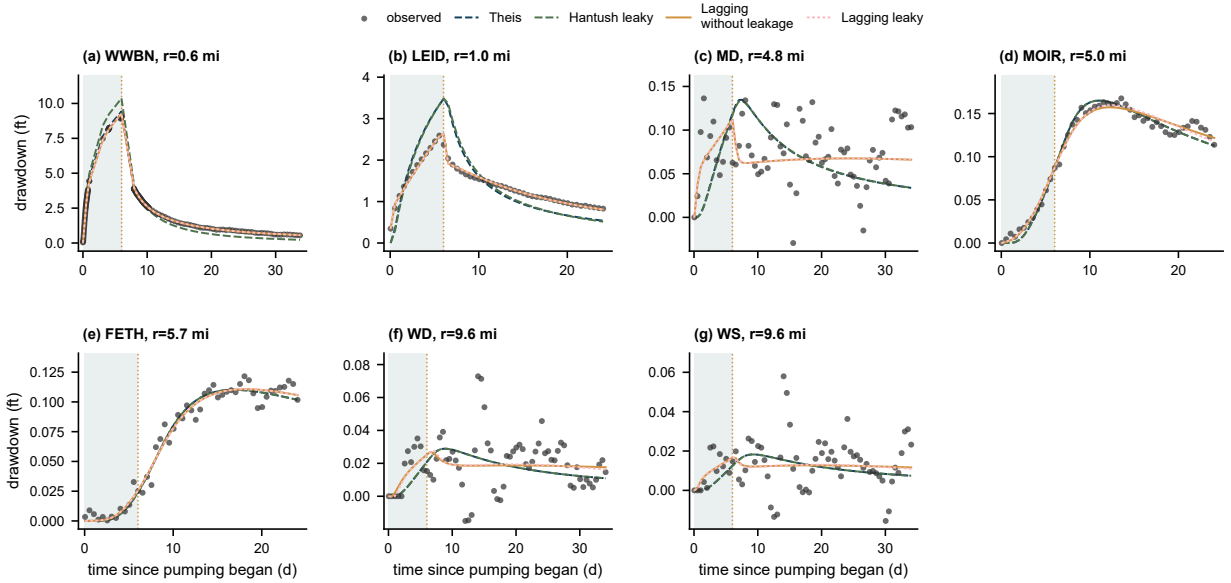


Figure 8: Lovelock Valley MWP2 observed drawdown and recovery responses with fitted curves under the four main interpretation models across seven selected observation wells by distance from Well 10, with measured responses as points, four model predictions as lines, the shaded interval as the extraction period, and the dotted line as shut-off.

Acknowledgments

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Computer Code Availability

Name of code/library: Aquifer-test transformation uncertainty benchmark workflow.

Developer and contact: Ying-Fan Lin, Department of Civil Engineering, Chung Yuan Christian University, yflin1110@cycu.edu.tw.

The source code is openly available under the MIT License at <https://github.com/aar246860/aquifer-test-tu-benchmark>. The workflow uses Python 3 and LaTeX. Python dependencies include NumPy, pandas, SciPy, scikit-learn, matplotlib, cmcrameri, FloPy, and Pillow. MODFLOW 6 is required only for rerunning the full numerical groundwater-flow simulations; the repository quick test does not rerun the 10,000 scenario benchmark. A lightweight reproducibility check can be run with `python examples/quick_test.py`. The larger generated benchmark and field-analysis tables are archived in HydroShare at <https://www.hydroshare.org/resource/3e8de7ff140c4ffab03163e215bb6634/>.

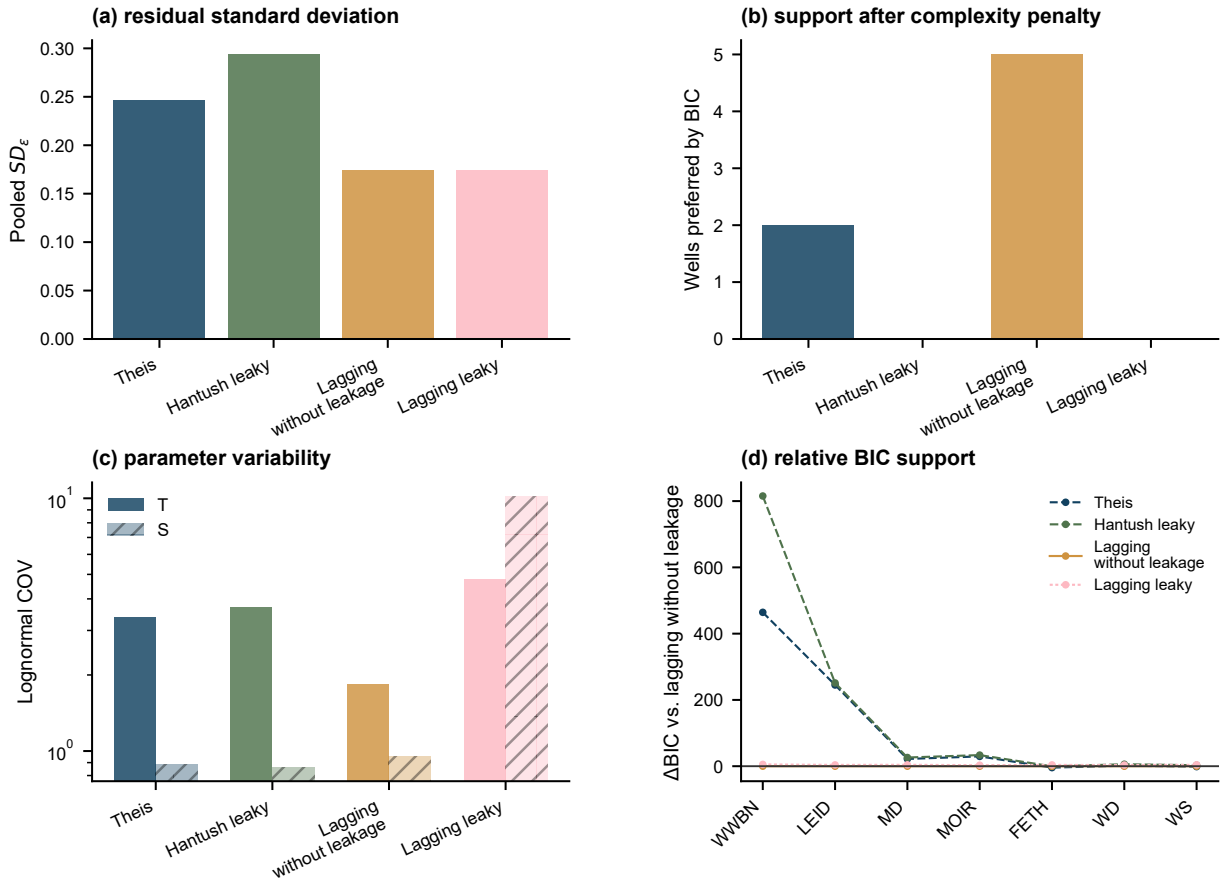


Figure 9: Lovelock Valley interpretation-model summary through (a) pooled log model factor dispersion; (b) BIC preference counts across the four main interpretation models; (c) lognormal parameter COV for fitted transmissivity and storage; and (d) relative BIC support against lagging Darcy without leakage.

References

- Anderson, M.P., Woessner, W.W., Hunt, R.J., 2015. Applied Groundwater Modeling: Simulation of Flow and Advective Transport. 2 ed., Academic Press.
- Bakker, M., Post, V., Langevin, C.D., Hughes, J.D., White, J.T., Starn, J.J., Fienen, M.N., 2016. Scripting MODFLOW model development using Python and FloPy. Groundwater 54, 733–739. doi:10.1111/gwat.12413.
- Barker, J.A., 1988. A generalized radial flow model for hydraulic tests in fractured rock. Water Resources Research 24, 1796–1804. doi:10.1029/WR024i010p01796.
- Bear, J., 1979. Hydraulics of Groundwater. McGraw-Hill, New York.
- Beckie, R., Harvey, C.F., 2002. What does a slug test measure: An investigation of instrument response and the effects of heterogeneity. Water Resources Research 38, 1290. doi:10.1029/2001WR001072.
- Boulton, N.S., 1954. The drawdown of the water-table under non-steady conditions near a pumped well in an unconfined formation. Proceedings of the Institution of Civil Engineers 3, 564–579. doi:10.1680/ipeds.1954.12586.
- Cho, H.J., Fiocco, Jr., R.J., Daly, M.H., 2004. Pumping test analysis in a fractured crystalline bedrock, in: Proceedings of the 2004 US EPA/National

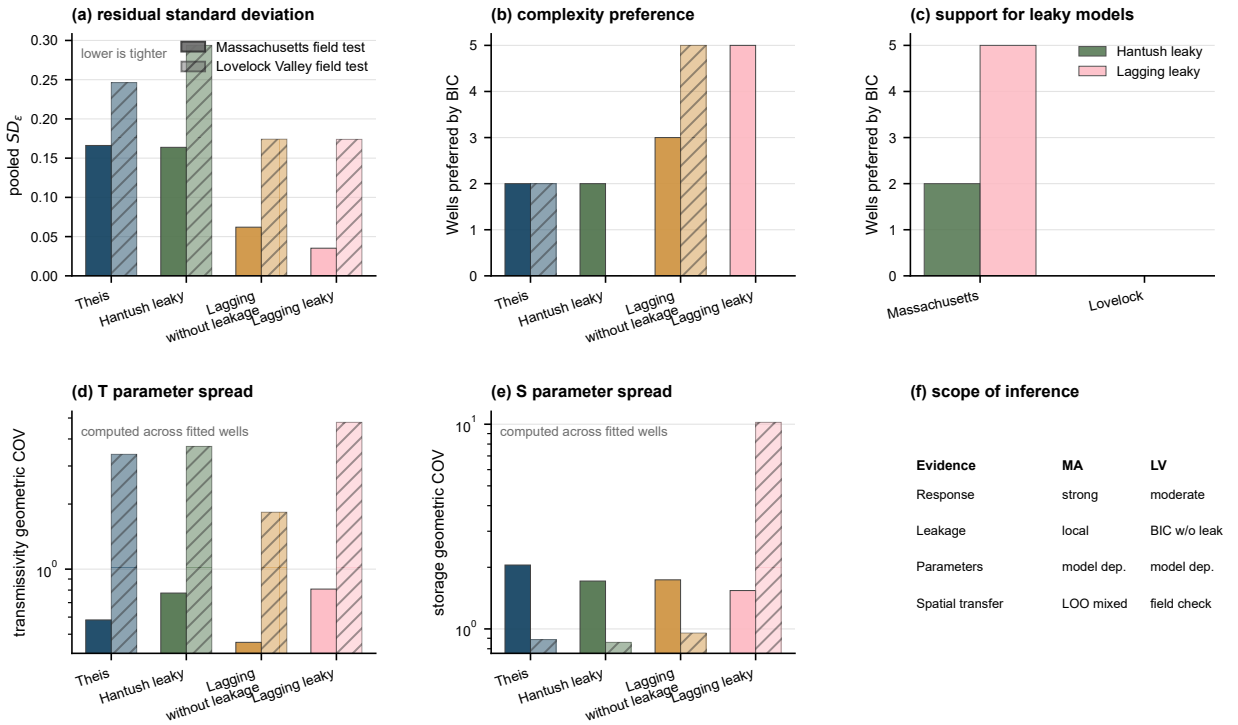


Figure 10: Validation synthesis from the Massachusetts fractured bedrock case and the Lovelock Valley basin-fill case through (a) pooled log model factor dispersion; (b) BIC preference counts by well; (c) support for leakage-inclusive models; (d) transmissivity COV; (e) storage COV; and (f) inference scope across response support, leakage support, parameter interpretation, and spatial transfer availability.

Ground Water Association Fractured Rock Conference: State of the Science and Measuring Success in Remediation, pp. 161–172.

Cho, H.J., Fiacco, Jr., R.J., Daly, M.H., 2008. Characterization of crystalline bedrock contaminated by dense nonaqueous liquid. *Ground Water Monitoring & Remediation* 28, 49–59.

Cooper, H.H., Jacob, C.E., 1946. A generalized graphical method for evaluating formation constants and summarizing well-field history. *Eos, Transactions American Geophysical Union* 27, 526–534. doi:10.1029/TR027i004p00526.

Doherty, J., 2015. Calibration and Uncertainty Analysis for Complex Environmental Models. *Watermark Numerical Computing*.

Freeze, R.A., Cherry, J.A., 1979. *Groundwater*. Prentice Hall, Englewood Cliffs, New Jersey.

Hantush, M.S., Jacob, C.E., 1955. Non-steady radial flow in an infinite leaky aquifer. *Eos, Transactions American Geophysical Union* 36, 95–100. doi:10.1029/TR036i001p00095.

Hoeting, J.A., Madigan, D., Raftery, A.E., Volinsky, C.T., 1999. Bayesian model averaging: A tutorial. *Statistical Science* 14, 382–417. doi:10.1214/ss/1009212519.

Langevin, C.D., Hughes, J.D., Banta, E.R., Niswonger, R.G., Panday, S., Provost, A.M., 2017. Documentation for the MODFLOW 6 groundwater flow model. *Techniques and Methods 6-A55*. U.S. Geological Survey. doi:10.3133/tm6A55.

Li, X., Tsai, F.T.C., 2009. Bayesian model averaging for groundwater head prediction and uncertainty analysis using multimodel and multimethod. *Water Resources Research* 45, W09403. doi:10.1029/2008WR007488.

Manewell, N., Doherty, J., Hayes, P., 2024. Translating pumping test data into groundwater model parameters: A workflow to reveal aquifer

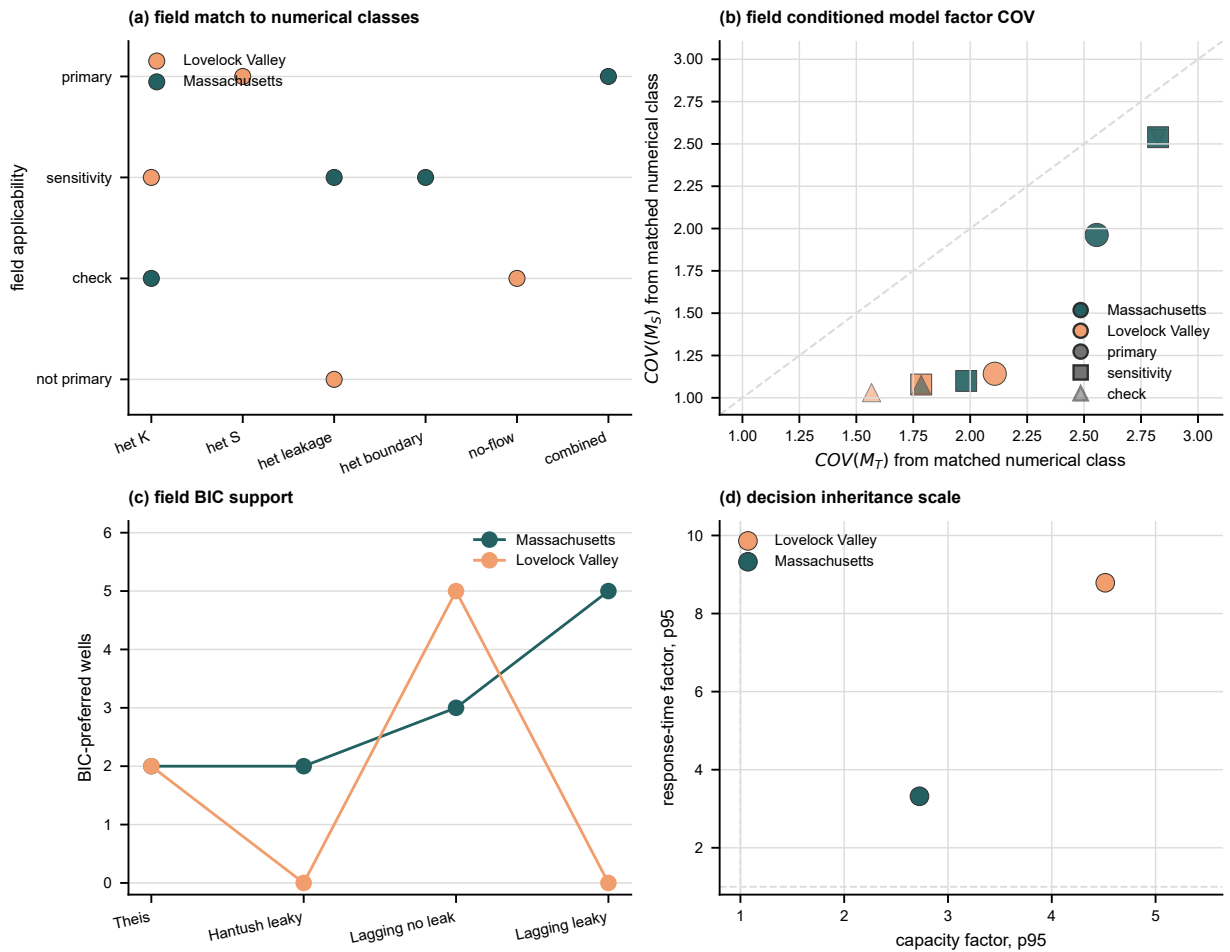


Figure 11: Field application of scenario-conditioned transformation model factors after field evidence-chain establishment through (a) assignment of the two field cases to numerical benchmark classes and applicability levels; (b) matched transmissivity and storage model factor COV values as field sensitivity brackets or conservative priors; (c) field BIC support for the four analytical interpretations; and (d) inherited 95th percentile capacity and response-time factors from the model-averaged decision diagnostic. The figure identifies where benchmark factors enter the field interpretation.

heterogeneities and implications in regional model parameterization. *Frontiers in Water* 5, 1334022. doi:10.3389/frwa.2023.1334022.

Manewell, N.M., Doherty, J., Hayes, P., 2023. Spatial averaging implied in aquifer test interpretation: The meaning of estimated hydraulic properties. *Frontiers in Earth Science* 10, 1079287. doi:10.3389/feart.2022.1079287.

Naderi, M., Gupta, H.V., 2023. On the requirements for inferring aquifer-scale T and S in heterogeneous confined aquifers. *Water Resources Research* 59, e2023WR034430. doi:10.1029/2023WR034430.

Nadler, C., 2020a. Analysis of aquifer framework and hydraulic properties of Lovelock Valley, Lovelock, Nevada. Open-File Report 2019-1133. U.S. Geological Survey. doi:10.3133/ofr20191133.

Nadler, C.A., 2020b. Supplemental data for analysis of aquifer framework and hydraulic properties of Lovelock Valley, Lovelock, NV. doi:10.5066/P9LIL7PZ.

Neuman, S.P., 1975. Analysis of pumping test data from anisotropic unconfined aquifers considering delayed gravity response. *Water Resources*

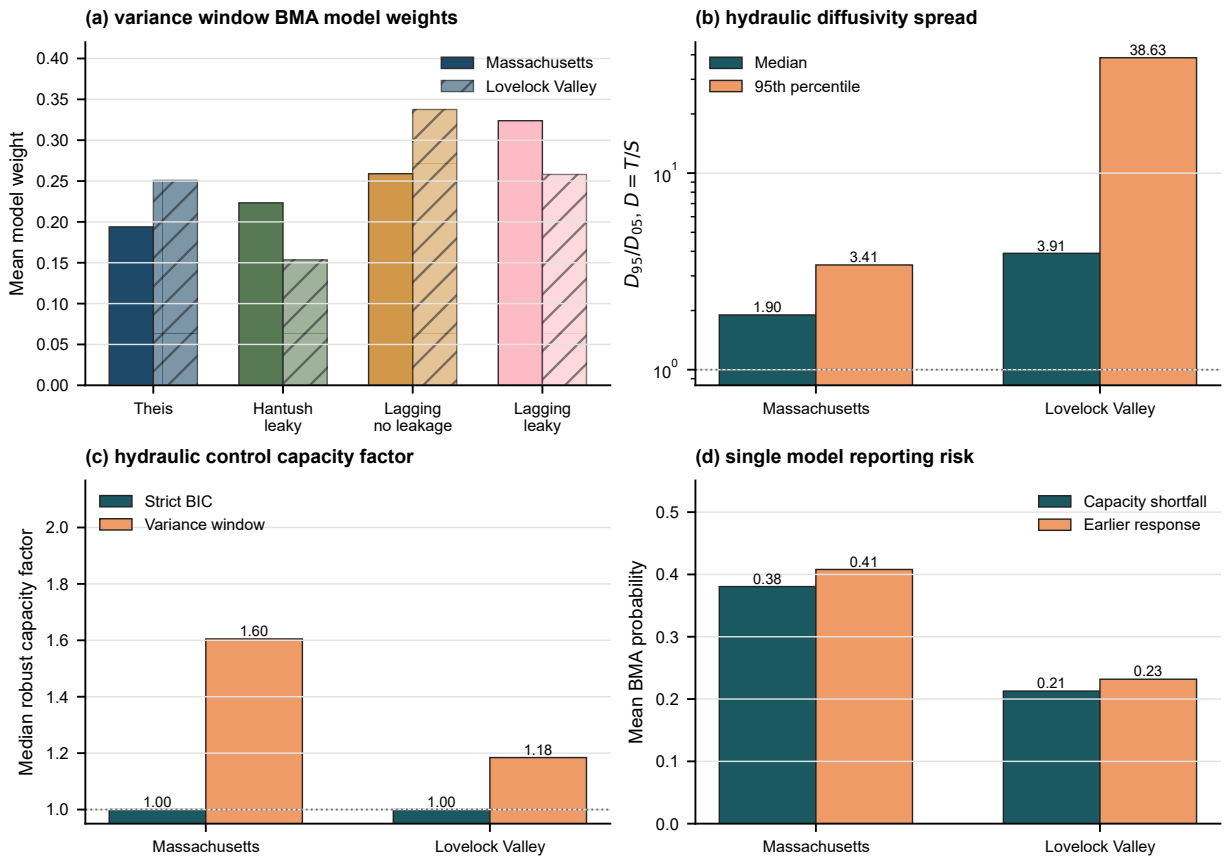


Figure 12: Model-averaged propagation of aquifer-test interpretation uncertainty to management-relevant hydraulic quantities through (a) mean variance-window BMA weights for the four analytical interpretations; (b) median and 95th percentile hydraulic diffusivity spread, expressed as D_{95}/D_{05} with $D = T/S$; (c) median robust hydraulic control capacity factor under strict BIC weights and variance-window weights; and (d) mean variance-window BMA probability of selected-interpretation underestimation of hydraulic control capacity or missed earlier hydraulic response.

Research 11, 329–342. doi:10.1029/WR011i002p00329.

Papadopoulos, I.S., Cooper, H.H., 1967. Drawdown in a well of large diameter. *Water Resources Research* 3, 241–244. doi:10.1029/WR003i001p00241.

Rojas, R., Feyen, L., Dassargues, A., 2008. Conceptual model uncertainty in groundwater modeling: Combining generalized likelihood uncertainty estimation and Bayesian model averaging. *Water Resources Research* 44, W12418. doi:10.1029/2008WR006908.

Stehfest, H., 1970. Algorithm 368: Numerical inversion of laplace transforms. *Communications of the ACM* 13, 47–49. doi:10.1145/361953.361969.

Theis, C.V., 1935. The relation between the lowering of the piezometric surface and the rate and duration of discharge of a well using ground-water storage. *Transactions, American Geophysical Union* 16, 519–524. doi:10.1029/TR016i002p00519.

Tsai, F.T.C., 2010. Bayesian model averaging assessment on groundwater management under model structure uncertainty. *Stochastic Environmental Research and Risk Assessment* 24, 845–861. doi:10.1007/s00477-010-0382-3.

Tsai, F.T.C., Elshall, A.S., 2013. Hierarchical Bayesian model averaging for hydrostratigraphic modeling: Uncertainty segregation and comparative evaluation. *Water Resources Research* 49, 5520–5536. doi:10.1002/wrcr.20428.